EECS 3311 Project Overview: AI Personal Finance Assistant

1.1 Project Description:

In a fast evolving society where financial pressures are constantly increasing, individuals face mounting challenges when it comes to saving and managing money. While many turn to bank loans as an immediate solution, this often creates a cycle of overwhelming debt, high interest rates, and long-term financial instability. In order to overcome these issues, the project proposes the idea of building an AI-powered financial assistant to serve the purpose of building a program that encourages saving and enables users to see the effects of their financial choices over the long term. The target users for this app will include individuals who are looking to improve their financial habits, manage their spending, build savings, and better understand the long term effects of their financial decisions. This includes young adults living on a tight monthly budget and individuals willing to export bank or credit card statements to gain useful insights.

The agent will use tools to retrieve and calculate over the user's real transaction data, remember user preferences, and explain relevant findings in plain language. The LLM will decide what to look up and how to explain the results while deterministic tools perform the actual database queries and calculations, so the numbers shown to the user are accurate.

By using Anthropic's Claude AI model, the agent will be able to perform tasks such as automatic categorization, detect unusual spending habits, and generate monthly based reports on user spending. When a user imports a bank or credit-card statement, the agent assigns each transaction to a category such as groceries, dining, rent, or transport. Furthermore, the agent analyzes the user's transaction history to find charges that are unusual for their category and then explains why each transaction was flagged so the user can identify if the transaction is expected or not. At the end of each month, the agent gathers the key figures including total income, spending, the breakdown by category, budget performance, and other transactions. Then, the agent turns the information into a readable summary that highlights what changed and where the user can save. Integrating AI is appropriate because financial data is messy, personal, and hard to decipher. An LLM can decipher the data and present it in a meaningful way that a user can understand.

The AI model will interact with the rest of the software system through an agent controller that connects Claude to the application's financial data, tools, and user interface. When a user performs an action through the GUI, such as importing a bank statement or asking a financial question, the request will be sent to the agent controller. The agent will determine what information and tools are required to complete the request and will then send the relevant context to the Claude AI model. The results from these tools will then be returned to the agent, which will provide the relevant information to Claude. Claude will interpret the results and generate an understandable response or report for the user.

1.2 Feature Specification:

F01	Importing data
Description	Loads transactions from a bank or credit-card statement file into the system.
User Interaction	In the GUI, the user will use the file button followed by choosing a file and then review a preview of the parsed rows then click confirm.
Input	Bank name → Account type → Upload CSV
Output	Transactions saved to the database and an import summary report showing the number of rows imported, duplicates skipped, and invalid rows rejected.
AI Involvement	Deterministic
Expected Workflow	1) The user selects a file. 2) The system selects the parser that matches the chosen bank's CSV layout 3) Each row is converted into a Transaction (date, amount, description, account). 4) Rows are validated and duplicates are detected. 5) A preview is shown. 6) On confirmation, the transactions are saved
Error/Alternative Cases	1) Invalid file type: System will prompt user for CSV 2) Empty file: If no text is detected, system will notify user 3) Duplicate rows: Notify user if two rows are identical

F02	Automatic Categorization
Description	Assigns every imported transaction to a dedicated category such as groceries, dining, rent, or transport. Each category will contain the name, total amount, and percentage to the user's total spending.
User Interaction	In the GUI, the user imports transactions and the system automatically assigns a category to each transaction. The user can review the assigned categories and manually change any incorrect category
Input	Transactions with descriptions and amounts, the list of allowed categories, and stored category rules
Output	Each transaction is assigned a category and saved to the database. Also, displays the data
AI Involvement	Hybrid because user uploads the CSV file and AI determines each transaction
Expected Workflow	1) Apply the stored rules to each transaction. 2) Collect the unmatched descriptions and send them to Claude with the list of allowed categories. 3) Parse Claude's structured response 4) Allow users to review. 5) Save.
Error/Alternative Cases	1) AI categorizes transaction into wrong type 2) The agent times out

F03	Manual Category Correction
Description	Lets the user override an AI-assigned category on any transaction and optionally save a rule so future matching transactions are categorized the same way
User Interaction	The user clicks a transaction row in the GUI → selects "Edit Category" → chooses from the allowed category list → optionally checks "Apply to similar transactions."
Input	Transaction ID, new category, optional rule flag
Output	Updated transaction in the database and stored category rule.
AI Involvement	Deterministic because the correction is user-driven
Expected Workflow	1) The user selects a transaction. 2) The system displays the current category and full category list. 3) The user picks a new category. 4) If "Apply" is checked, the system creates a CategoryRule from the description pattern. 5) The system updates the transaction and notifies the dashboard.
Error/Alternative Cases	1) Rule conflicts with existing rule. In this case, system warns user and asks to replace or keep both 2) The user cancels mid-edit: No changes are saved.

F04	Set Budget
Description	Allows the user to set a monthly spending limit per category.
User Interaction	Budget panel in the GUI → "New Budget" → pick month, category, and amount → save.
Input	Month, category, budget amount.
Output	Budget record saved and displayed on the dashboard.
AI Involvement	Deterministic because the budget is set based on user input
Expected Workflow	1) The user opens the budget panel. 2) The system lists categories. 3) The user enters a month, category, and amount. 4) The system validates the amount (positive number, reasonable range). 5) Budget is saved via BudgetRepository. 6) Dashboard updates
Error/Alternative Cases	1) If negative or zero amount, then the system rejects and prompts again 2) The user already has a budget for that month or category in which case the system asks to be overwritten or cancelled.

F05	Budget Alerts
Description	Compares actual spending against each budget and alerts the user when spending nears or exceeds a limit.
User Interaction	User interacts with an alert panel with a notification icon in the GUI that updates as transactions change
Input	Budget records, categorized transactions for the month, and near threshold limit
Output	Alert messages to users containing the amount of money spent for category and remaining.
AI Involvement	Deterministic because the alert will be based on the pre-set limit done by the user
Expected Workflow	1) A transaction is added or changed. 2) The affected category total is recalculated. 3) The system checks the thresholds. 4) If a threshold is crossed for the first time that month, an alert is created and displayed.
Error/Alternative Cases	1) Division by zero: System shows "no budget set" instead of a percentage 2) No budgets defined: System shows a prompt to create budgets.

F06	Manage Subscriptions
Description	Recognizes repeating charges which can include rent, entertainment, or gym and tracks them as subscriptions.
User Interaction	Subscriptions tab from GUI where users can confirm, rename, or mark a subscription.
Input	Transaction history as CSV
Output	Subscription name, total amount, next expected date, and price increases
AI Involvement	This task will involve deterministic logic to detect the recurring payments by grouping transactions by merchant and checking for similar amounts at regular intervals. The AI will help by offering alternatives. For instance, if a user has Spotify and Apple Music, the model could highlight the overlap between them.
Expected Workflow	1) A user uploads CSV transaction history 2) The data is processed using AI where subscriptions are categorized by service. For example, Spotify and Apple Music subscription are both music streaming platforms 3) User goes to subscription tab to view name, total amount, expected date, and price increases 4) The AI agent will show a message to show any overlap
Error/Alternative Cases	1) Limited transaction history: if the CSV is provided for one month, detecting a subscription would not be possible as nothing repeats. Thus, the system notifies the user that more history is needed to detect subscriptions. 2) No transaction history: the system shows a message asking the user to upload a CSV again

F07	Natural ChatBox with Agent
Description	This feature allows the user to freely communicate with the agent for financial advice regarding such as spending, budgets, subscriptions, and goals. The agent will use the user's transaction history to provide a personalized response.
User Interaction	In the GUI, an AI chat panel will be provided where the user types a question and receives an answer through a message.
Input	An open-ended question from the user about their financial status or goals
Output	A natural-language answer based on user transactions
AI Involvement	This task will be AI-based
Expected Workflow	1) The controller sends the question and memory to the agent. 2) Claude chooses a tool to call 3) The tool runs and returns data. 4) Claude writes the answer from the tool results. 5) The exchange is saved in conversation memory
Error/Alternative Cases	1) If the agent is unavailable, use a timer and send out a message that AI-system is down

F08	Savings Goal Management
Description	Allows the user to set savings goals, record contributions, and see progress and the monthly amount needed.
User Interaction	In the GUI, there will be a Goals tab where users can then enter a New Goal with attributes such as name, target, deadline, and contribute button.
Input	The parameters the user will enter are goal name, target amount, deadline, and contributions.
Output	The output will contain progress percentage, required monthly contribution, determine if the user is on track to meet the goal, and an AI message that explains the user's progress.
AI Involvement	1) The user creates a goal. 2) The system calculates the monthly amount needed. 3) Contributions update the progress. 4) The system estimates the completion date from recent savings. 5) If the goal is behind, the agent suggests adjustments.
Expected Workflow	1) The user creates a goal with a name, target amount, and deadline. 2) The system calculates the monthly amount needed. 3) The user adds contributions, which update the progress. 4) The system estimates the completion date and shows the progress percentage, required monthly contribution, and whether the user is on track. 5) If the goal is behind, the agent suggests adjustments to get back on track. 6) When the target is reached, the goal is

	marked as completed.
Error/Alternative Cases	1) If the goal is already completed, the system will mark it as done and stop displaying the required contribution 2) Deadline target 0 or less, the system will flag it and show an error

F09	Monthly Report Generation
Description	This will produce a summary of a selected month, including total income, total spending, the breakdown by category, budget performance, and where the user can save.
User Interaction	In the GUI, the user opens the reports tab, selects a month, and clicks generate. The report is displayed on screen
Input	The parameters the user will enter are the selected month, the categorized transactions for that month, budget records, and detected subscriptions.
Output	A monthly report containing key figures and a written summary with advice for saving opportunities
AI Involvement	This process will be hybrid. The feature will calculate the figures deterministically while the agent writes the summary from those figures
Expected Workflow	1) The user selects a month and clicks Generate. 2) The system gathers the month's totals, category breakdown, and budget status. 3) It collects data such as unusual transactions and subscription changes. 5) The facts are sent to the agent who writes the summary. 6) The report is saved and displayed.

Error/Alternative Cases	1) No transactions in the selected month. In this case, the system displays a warning to user

F10	Report Export
Description	This feature enables the user to export reports such as monthly reports, transactions, and budget status to a file the user can save or share.
User Interaction	In the GUI, there will be an exports tab where users can choose the report type, month, and format, then click Save
Input	The parameters the user will enter are report type, period, format, and destination path.
Output	A file will be produced on the computer containing the selected report
AI Involvement	This process will be deterministic
Expected Workflow	1) The user selects the report options. 2) The system collects the data. 3) The exporter for the chosen format is selected. 4) The file is written and its location is shown
Error/Alternative Cases	1) Path error (invalid or does not exist). In this case, show an error and ask for a new path 2) If file already exists, prompt user to overwrite